

Clearcutting forestry and Eurasian boreal forest grouse: Long-term monitoring of sympatric capercaillie *Tetrao urogallus* and black grouse *T. tetrix* reveals unexpected effects on their population performances

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ABSTRACT

Along the succession gradient of the boreal forest ecosystem, black grouse *Tetrao tetrix* inhabits the early and capercaillie *Tetrao urogallus* the latest stages. When converting old forest to clearcuts and plantations, commercial forestry has therefore been assumed to affect capercaillie negatively and to be favourable to black grouse. During a 30-year period (1979–2008) we monitored sympatric populations of the two species in a forest in southeast Norway based on annual spring and autumn censuses and radio-marked birds. During this period, the proportion of old, semi-natural forest was halved and clearcuts and young plantations increased accordingly. The grouse populations did not change as predicted. While the trend in August numbers of adult black grouse declined, males more than females, abundance of adult capercaillie remained unchanged. Number of males at leks showed similar patterns. Equally surprising, breeding success (number of chicks per female in August) of both species increased, thus indicating that the populations were regulated more by variation in adult survivorship than by recruitment of young birds. No correlations were found with changing climatic factors (precipitation and temperatures in winter and spring, snow depth and time of snow melt), except that year-to-year breeding success was positively correlated with minimum temperatures during 2 weeks posthatch. The results are explained by a combination of more flexible habitat selection than previously assumed and a changing predator regime: In the early period, nearly all capercaillie leks were located in old, semi-natural forest, but as plantations grew older (>30 years), new leks were established there. Similarly, while young capercaillie broods used old semi-natural forest almost exclusively when the study started, they frequently used middle-aged plantations, especially those with a ground cover of bilberry *Vaccinium myrtillus*, when these became common in later years. The increasing breeding success could largely be explained by more females rearing chicks successfully, presumably due to a marked decline in the main nest predator, the red fox *Vulpes vulpes*. A practice of thinning of the old, semi-natural forest some years prior to final harvesting probably facilitated predation of black grouse by goshawks *Accipiter gentilis*. Contrary to many beliefs, our results indicate that both capercaillie and black grouse are quite tolerant to changes in forest management regimes. In our study, numerical and functional responses of predators (mainly red fox and goshawk) apparently played a more important role in regulating grouse numbers than habitat factors per se.

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1. Introduction

In Eurasian boreal forests, the lekking and highly sexually dimorphic black grouse *Tetrao tetrix* and larger capercaillie *Tetrao urogallus* are known to occupy opposite stages along the successional gradient in the landscape; the smaller black grouse (BG) inhabits the early stages and capercaillie (CAP) the later seral stages

(Seiskari, 1962; Klaus et al., 1989, 1990; Swenson and Angelstam, 1993). Both are ground nesters, with the ca. 0.9 kg BG female laying between 8 and 9 eggs, and the ca. 1.9 kg CAP ca. 7 eggs. Mating takes place in late April–early May, and eggs hatch during the first two weeks in June.

Until the mid 1900s, spruce and pine forests in Fennoscandia were harvested by various forms of selective cuttings. Smaller trees were used for fuelwood and farming materials, whereas larger trees were cut mainly for commercial purposes. Under this management regime, the composition of residual trees of various sizes resembled the pre-cut condition; only tree density was reduced. When

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clearcutting became the main harvesting and regeneration practice in Fennoscandia in the 1950s, the forest landscape was rapidly fragmented into a checkerboard pattern; the proportion of semi-natural, old forest decreased while the proportion of clearcuts and even-aged, young plantations increased. Today – after 60 years of clearcutting forestry – the old semi-natural forest occurs as patches embedded in a matrix of clearcuts and even-aged, culturally regenerated stands.

In all the three Nordic countries, CAP declined markedly after the early 1960s (Rajala, 1979; Sørensen, 1977; Wegge and Grasaas, 1977; Marcström, 1979; Wegge, 1979; Linden and Rajala, 1981; Helle et al., 2003; Lakka and Kouki, 2009; Sirkiä et al., 2010). Fragmentation and loss of old forest was considered the main reason for this, as inferred from several habitat studies conducted in the boreal zone (e.g. Rolstad, 1989; Wegge et al., 1992; Sjöberg, 1996; Kurki et al., 2000) and in Continental Europe (Klaus, 1991; Saniga, 2003) and Scotland (Moss and Oswald, 1985). Although BG also declined locally, in the boreal region their numbers appeared far less affected by the changes in forest composition induced by logging (Lindström et al., 1998; Lindén and Helle, 2003; Angelstam, 2004; Storch, 2007).

In Norway, a pronounced decline in CAP in the 1970s led to a comprehensive research project in the early 1980s, aimed at identifying how the current logging practices affected the populations. In the Varaldskogen study area, various short-term studies were conducted, and sympatric populations of the two species have been monitored during a period of 30 years. Here, we report how the populations changed during this period and compare the population trends with the concurrent changes in forest composition. Based on the two species' habitat niches, predictions were that (1) with the loss and fragmentation of old forest, numbers of CAP should decline, and (2) with the "opening up" of the landscape and creation of more early successional habitats, numbers of BG should increase. Because the populations developed and performed differently than expected, we examine other factors than forest age structure that are assumed to affect the population performance of forest grouse (climate and predation). Changes in habitat use were examined by the monitoring of radio-marked birds.

2. Study area

Located next to the Swedish border in southeast Norway (60°10'N, 12°30'E), the study area covers ca. 100 km², of which the eastern part of the Varald State Forest (ca. 40 km²) constituted an intensively monitored core area. The gently undulating terrain between 200 and 400 m a.s.l. consists of Norway spruce *Picea abies* (40% of standing volume) and Scots pine *Pinus silvestris* (58%) forests interspersed with scattered birch *Betula* spp. (2%) and aspen *Populus tremula*. Black grouse (BG) and capercaillie (CAP) occur at moderately high patchily distributed hazel grouse *Bonasa bonasia*. Main predators are red fox *Vulpes vulpes*, pine marten *Martes martes* and goshawk *Accipiter gentilis*; other raptors are seasonal visitors or quite rare. During the mid-1980s, a regional outbreak of sarcoptic mange *Sarcoptes scabiei* disease markedly reduced the fox population (Lindström et al., 1994; Smedshaug et al., 1999), from which it later did not fully recover. Wolf *Canis lupus* has recently become a regular visitor, while lynx *Lynx lynx* occur occasionally. While BG was hunted each year, hunting of CAP was prohibited until 1995. However, mortality from hunting was low, rarely exceeding 5% of the number of birds of each species in autumn (unpubl.).

Contiguous with mixed conifer forests on all sides – only interspersed with small patches of farmlands and settlements – the forest has been subjected to timber harvesting for several centuries. In the 1950s, clear-cutting became the main harvesting regime, replacing the traditional high-grading and selection methods. Cut-

ting blocks were initially rather large (>50 ha in 1960–1970s), but have successively been reduced in size, today rarely exceeding 20 ha. However, because they have been added to recently cut blocks, the total landscape with rejuvenated forests now covers larger tracts.

Following clearfelling, spruce forests were replanted, whereas pine forests were mainly rejuvenated by leaving widely spaced seed trees for naturally regeneration. Up till the late 1980s, most regenerated stands were thinned and cleared of competing broadleaf regrowth, thereby producing more or less pure, even-aged conifer stands. In later years, more hardwood trees were allowed in plantations. In the new pine stands on better sites, spruce often became established along with the pine, thus forming more stratified, mixed regeneration stands. Nevertheless, the young and middle-aged forests that were created by silvicultural practices were structurally less variable, especially in the vertical plane, than naturally regenerated forests.

In the older, semi-natural forest, ericaceous shrubs dominate the field layer, heather *Calluna vulgaris* and cowberry *Vaccinium vitis-idaea* in the less productive pine forest and forested peat bogs and bilberry *Vaccinium myrtillus* in the spruce-dominated forest on mesic sites. Following clearfelling and replanting, bilberry is usually replaced by a mixture of forbs and graminoids, mainly *Deschampsia* and *Calamagrostis* spp. Logging of pine forests usually causes less pronounced changes in the field layer, as here heather and cowberry *Vaccinium vitis-idaea* tend to maintain their cover. Oligotrophic peat bogs cover ca. 10% of the landscape, consisting of graminoid-dominated central parts surrounded by an ecotonal belt of mixed ericaceous shrubs and open pine forest. Where the bogs are small and the terrain is flat, the vegetation becomes dominated by scattered pine and spruce with a mixture of *Vaccinium* spp. and bog cottongrass *Eriophorum vaginatum* in the field layer.

3. Methods

3.1. Forest composition

Changes in forest composition were extracted from forest inventories (every 10th year) and statistical records maintained by the state forest office. These records contain information on age class distribution by species and site quality, as well as implemented silvicultural practices. All data are presented on a forest stand basis. We transferred the available data to digitized maps and measured spatial distributions and fragmentation pattern in ArcView/GIS 3.3 software (ESRI Inc., Redlands, CA, USA). The fragmentation pattern was assessed using an index: $P/(4A^{0.5})$, where P is the perimeter and A is the area of the age-class category of interest. This ratio – the observed length of edge to the length of edge if the age-class category been clumped in one square patch – is independent of the total percentage coverage. Because we were primarily interested in assessing the effect of changes in the forest landscape on capercaillie, fragmentation was measured for all semi-natural forest patches >70 years, and for all stands >40 years (which included older plantations).

3.2. Bird population trends

3.2.1. Spring numbers

Distribution of leks and number of attending birds were monitored regularly, but reliable numbers were not obtained on each lek every year. That applied especially to the number of females and yearling CAP males, which are difficult to count and often visit more than one lek during the mating season (Wegge and Rolstad, 1986; Storch, 1997). Also, CAP leks shifted location as a consequence of logging (Rolstad and Wegge, 1989), and in some cases we did not

discover the new lek sites the same year they were established. Hence, when assessing changes in spring numbers, we only used lek data of males recorded during two shorter periods at the beginning (1980–1983) and end (2005–2008) when all leks within a slightly expanded core area of ca. 50 km² were monitored intensively. Presumably, numbers of males at CAP leks were below carrying capacity set by habitat and territorial behaviour (Wegge and Rolstad, 1986). Thus, we assume that numbers of attending birds reflected the general changes in the breeding populations. The distribution of BG leks and number of attending males were recorded and compared for the same periods as CAP (Rolstad et al., 2009).

3.2.2. Autumn numbers

We sampled the autumn populations in August with pointing dogs. The 40-km² core area was divided into forty-two 1.1–2.8-km² blocks, which were searched by 12–20 persons with well-trained dogs. Besides species, sex and age, they also recorded time and flight direction of flushed birds in order to minimize double-counting of birds that were flushed into adjacent blocks being sampled simultaneously. The man–dog teams did not sample along prefixed transect lines; instead they were instructed to sample the blocks intensively and attempt to find as many birds as possible. Thus, the collected data could not be used for absolute density estimation; instead they were converted to an index of abundance of number of individuals observed per 10 h. Preferably, all bird censuses should have been conducted under favourable, or constant, weather conditions for the dogs. However, only to a limited extent could the dates of the annual censuses be adjusted to achieve this. Hence, sampling variation due to variable weather conditions was unavoidable and is inherent in the final data set. An average of 132 h of census work was conducted each year, covering ca 2/3 of the main study area.

During the census, number of adults by sex and number of chicks in the broods were recorded. Whenever possible, the sex of the chicks in CAP broods was also noted; this was not attempted in BG broods as plumage differentiation among chicks in this species is not evident before quite late in August.

3.3. Habitat use and mortality

Radio telemetry provided information on seasonal habitat use at different time periods. During 1980–1989 and 1996–2003, we studied a total of 229 and 33 adult radio-marked CAP and BG, respectively. Here, we extract relevant results from those studies. During 2005–2007, another 65 BG were radio-marked. Besides obtaining information on habitat use, we used the radio-marked birds to compare rates and causes of mortality in the early and late part of the 30-year monitoring period. From the August censuses, yearly loss of birds was calculated as: $1 - (Ad_{year\ 2} / (Ad + Juv)_{year\ 1})$. The trend in the abundance of red fox was inferred from hunting statistics for the whole Varald State Forest (ca. 80 km²). This was supplemented by systematic counts of number of fox tracks crossing forest roads and snowmobile trails after snowfall in intermittent years, reported as tracks per 24 h per 10 km.

3.4. Weather

Due to climate change, a gradual change in weather factors may have affected the grouse populations, particularly during the reproduction phase (Moss, 1986; Moss et al., 2001; Ludwig et al., 2006). We therefore examined if there had been a directional change in weather during the 30-year period, which correlated with changes in bird breeding success and abundance. From the nearest meteorological stations we downloaded various weather variables (precipitation and temperatures in winter and spring/early summer, snow depth and date of snow melt, NMI, 2009) and analysed patterns of long-term trends during the 30-year monitoring period.

3.5. Analyses

Statistical significance of trends in bird abundance in August was assessed by two-tailed testing of OLS regression slopes using Student's *t*-statistics. In case of autocorrelation, effective sample size (N_{eff}) and adjusted standard errors (SE_{adj}) were calculated from the lag-1 autocorrelation (r_1) of the residual deviances according to Bartlett (1935) and Santer et al. (2000), as modified by Bretherton et al. (1999) and Thompson and Solomon (2005) to be less conservative at lower serial dependencies: $N_{eff} = N(1 - r_1^2)/(1 + r_1^2)$.

Because recruitment of chicks in CAP was female-biased (see Section 4 or 5), we adjusted the sex composition of the number of recorded chicks in August to a flat proportion of 64% females and 36% males throughout the study period. In BG, autumn chicks also tend to be female-biased, although less pronounced compared to CAP (Hörnfeldt et al., 2001). For this species, we used a male:female ratio of 45:55 in our trend analyses.

Mortality rates of adult radio-marked birds were estimated from proportion of marked birds known to have died within monthly intervals. Because birds were tracked irregularly – some at intervals >1 month – date of death could not be determined accurately. Hence, to compare mortality rates between the early and late parts of the 30-year monitoring period, the data were standardized and analysed as: percent mortality = $1 - (sb/tb)100$, where *sb* = number of birds known to have survived during the month, and *tb* = total number of birds alive at the start of the month. Because all birds survived an unknown number of days after the last month they were known to have been alive, this formula overestimates mortality. In order to correct for this, individual length of survival was extended as follows: Since estimated dates of death were randomly distributed within a month, each bird was assumed to have been alive for half of a month succeeding the latest month when recorded alive. Twelve birds, which we lost radio contact with before end of prescribed battery lifetime, were omitted from the data set. Owing to small sample sizes when broken down to sex and time periods, the data were analysed calculating monthly proportions and tested by the Fisher's Exact Test (FET).

Habitat use of radio-marked birds was examined by Chi-Square with Bonferroni adjustments of the alpha level and illustrated by Ivlev's selectivity index (adults): $D = (u - a)/(u + a - 2ua)$, where *u* is the proportion of a given forest age class used by the birds and *a* is its weighed proportion available to the birds (Ivlev, 1961; modified by Jacobs, 1974). The index ranges from –1 (total avoidance) through 0 (no selection) to +1 (maximum positive selection).

4. Results

4.1. Changes in the forest landscape

The change in forest age composition is shown in Figs. 1 and 2. From 1979 to 2008, the proportion of old forest (>70 years old) declined from 51% to 22%, constituting a yearly loss of 1%. As a result of intensive logging in the 1970s and early 1980s, the fragmentation pattern became increasingly fine-grained with the remnant old forest being split into smaller and smaller patches embedded in a matrix of younger stands (Figs. 1 and 2). The proportion of middle-aged plantations (21–40 yrs) increased from a low 8% in 1979 to 45% in 2000 (Fig. 2a). Similarly, the proportion of older plantations increased markedly during the last decade. As the earliest plantations grew older, they merged with the remnant old forest patches to create larger blocks of older, mixed-aged forest. Thus, during the last decade, stands >40 years became more coarse-grained and less fragmented then in the 1990s (Fig. 2b).

Because clearfelling and replanting started at lower elevations on the better, spruce-dominated sites, spruce plantations were

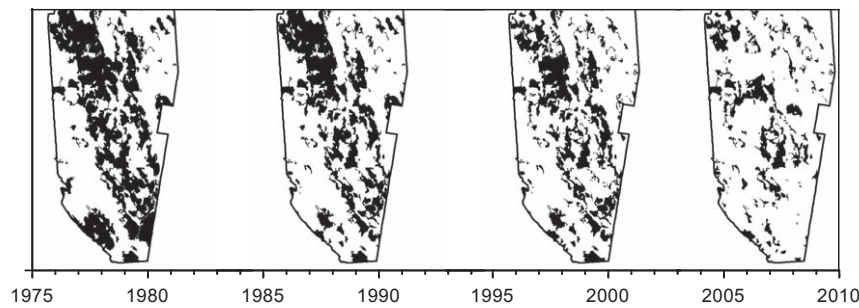


Fig. 1. Increasing loss and fragmentation of old, semi-natural forest >70 years (black) in the southern part (ca. 30 km²) of the Varaldskogen study area during 1979–2008.

generally older than the pine-regenerated stands on drier sites at higher elevations. In spite of cleaning operations to remove competing broadleaf saplings in the new plantations, the proportion of birch did not decline; both total standing volume and number of standing trees (≥ 20 cm dbh) increased during the study period.

4.2. Population trends – adult birds

Between 200 and 300 birds were recorded during the August census each year. Both species fluctuated markedly, both from year to year and on a decadal basis. Surprisingly and contrary to expectation, the long-term trend line of number of adult CAP in August did not decline ($b = -0.015$, $t_{\text{adj}} = -0.35$, $N_{\text{eff}} = 13$, $P = 0.73$), and that of BG grouse did not increase; instead it decreased during the 30-year period ($b = -0.129$, $t_{\text{adj}} = -3.39$, $N_{\text{eff}} = 27$, $P = 0.002$), Fig. 3a). This caused the BG/CAP ratio to decrease (ln ratio: $b = -0.025$, $t_{\text{adj}} = -3.23$, $N_{\text{eff}} = 19$, $P = 0.005$), from being almost two times in favour of BG prior to 1985 to a slight tendency (12%) in favour of CAP after 2000 (Fig. 3a). Populations of both species were female-biased. The more pronounced decline among males in BG led to

an increasing female skew in sex ratio in this species (ln ratio: $b = 0.0195$, $t_{\text{adj}} = 3.17$, $N_{\text{eff}} = 28$, $P = 0.004$, Fig. 3b). In CAP, the pattern was opposite: females declined slightly while males showed a small increasing trend, combined leading to a decreased female skew in this species (ln ratio: $b = -0.0148$, $t_{\text{adj}} = -2.67$, $N_{\text{eff}} = 22$, $P = 0.01$, Fig. 3b).

Number and distribution of CAP leks changed during the period. Starting with 10 leks in the early 1980s, 5 new leks – most of them

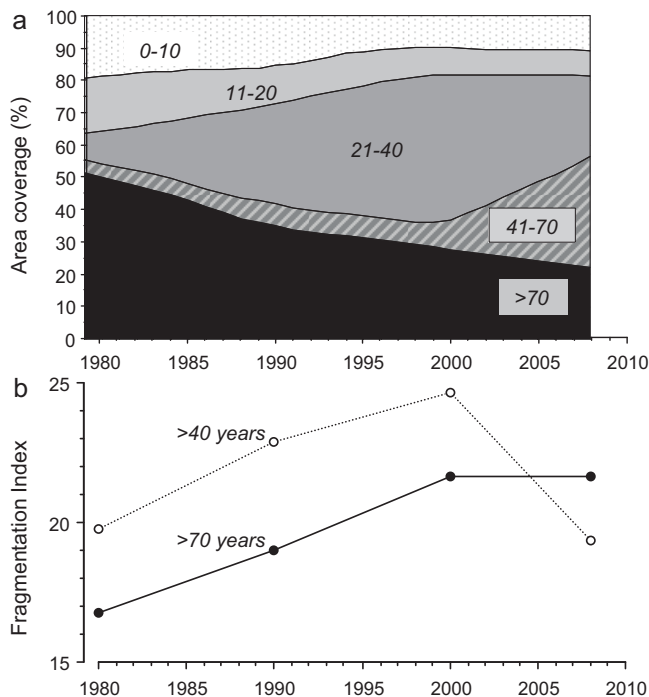


Fig. 2. Changes in (a) age composition and (b) fragmentation pattern of the forest in the Varaldskogen study area. The fragmentation index is the ratio of the observed length of edge of the age class category to the length of edge of the respective age-class if it was clumped in one square patch, i.e. the ratio of observed versus minimum fragmentation.

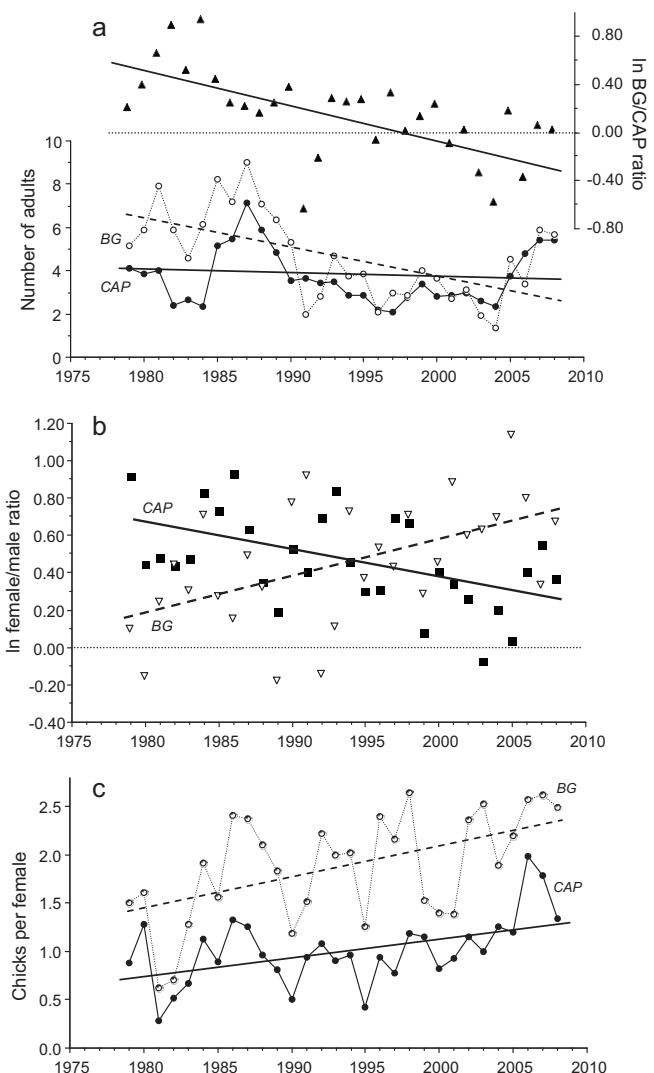


Fig. 3. Trends in (a) August numbers of adult capercaillie and black grouse and their ratios, (b) adult sex ratios of each species in August, and (c) breeding success (number of chicks/adult female in August) of each species.

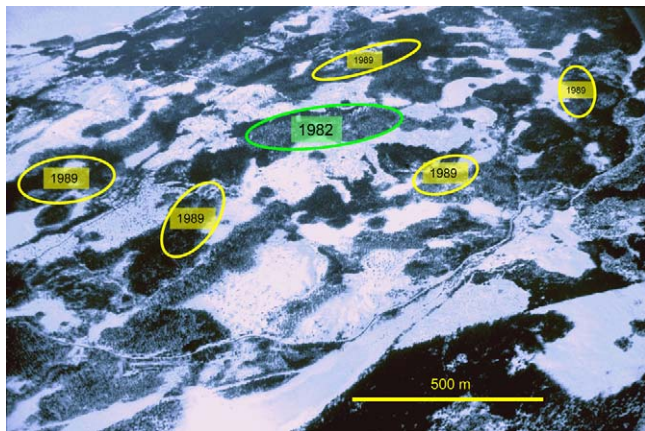


Fig. 4. The effect of logging on capercaillie leks: Following logging operations in 1985–1987 around an originally large lek (1982, green), the lek was abandoned and new, small leks (1989, yellow) were established – most of them temporarily – in nearby blocks of older forest. Old, semi-natural forest stands show up dark before thinning and grey after thinning. White patches are new clearcuts, very young plantations and frozen lakes (Photo taken February 1988). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of the article.)

Table 1

Number of leks and mean number of attending males within 50-km² during early (1980–1983) and late (2005–2008) parts of the 30-year monitoring period, Varaldskogen, Norway.

	Capercaillie ^a		Black grouse ^b	
	Early	Late	Early	Late
Number of leks	10	12	17	14
Number of males/lek	4.4	4.2	7.2	7.6
Total males/km ² /year	0.9	1.0	2.4	2.1

Source: Rolstad et al. (2009).

^a Age > 23 months.

^b Age > 11 months.

active only for a few years – were established during the latter part of that decade and the first half of the 1990s, during a period when fragmentation of old forest was quite dramatic. It was partly a result of logging most of the older forest around one large lek, which then ceased to exist (Fig. 4).

In general, number of birds at leks followed the trend in August abundances. Within the expanded core area (50 km²), a, number of CAP leks increased from 10 in 1980–1983 to 12 in 2005–2008. With number of attending birds per lek decreasing only slightly, this resulted in a slightly higher spring density of adult males in the latter period (Table 1). Seventeen BG leks were active within the core area when the study started. Two leks subsequently ceased to exist, presumably due to growing surrounding plantations eventually “choking” the display arenas, and 1 small lek was abandoned. With fewer leks but slightly more males/lek in the last period, the number of lekking BG males decreased from 2.4 to 2.1 per km², or ca. 13% (Table 1).

Table 2

Trends in weather data from Varaldskogen 1979–2008.

Variable	Mean	Min.	Max.	Range	Trend	Slope	t_{adj}^a	P-value	N_{eff}
Snow depth _{Max} (cm)	42	0	108	108	−47	−1.478	−2.60	0.015	27
Date of snow-free ground ^b	17	−8	38	46	−21	−0.695	−2.98	0.007	26
Winter $T_{Average}$ (°C)	−4.6	−10.4	−0.9	9.4	3.8	0.135	2.62	0.014	28
April 10–30 $T_{Average}$ (°C)	5.1	1.5	7.8	6.3	2.5	0.094	3.57	0.001	28
June $T_{Minimum}$ (°C)	8.6	6.2	10.4	4.2	1.3	0.043	2.33	0.028	28

^a Adjusted for autocorrelation according to effective sample size (N_{eff}) and effective SE (see Section 3).

^b Days after April 1.

4.2.1. Modifying factors–adults

The mean winter temperature (December–March) increased by 3.8 degrees, and snow depth decreased significantly during the 30-year study period (Table 2). At a first look, this suggested that the declining trend in BG could be related to a trend in winter weather. However, year-to-year variation in annual adult mortality was not correlated with winter temperature in either of the two sexes ($R_M^2 = 0.011$, $R_F^2 = 0.007$, $N = 29$, $P > 0.20$).

Mortality rates of radio-marked adult birds are shown in Table 3. In CAP, annual mortality appeared to be higher during 1996–2003 than earlier, but differences were not significant ($P > 0.32$, each sex, Fisher's Exact Test (FET)). In both species, there were tendencies for higher mortality in May–June during 2005–2008 than in the 1980s (BG sexes combined: $P = 0.062$, males only: $P = 0.183$; CAP sexes combined: $P = 0.076$, females only: $P = 0.152$, FET). In CAP, no differences were detected in either sex in the main four seasons ($P > 0.38$, FET, each sex and season). Predation by goshawk was by far the most important cause of mortality of BG and CAP females (Table 3).

4.3. Population trends – recruitment of young birds

Equally surprising, breeding success (chicks per female in August) of both species increased steadily throughout the period, CAP with 87% ($b = 0.0207$, $t_{adj} = 2.48$, $N_{eff} = 20$, $P = 0.023$) and BG with 74% ($b = 0.0353$, $t_{adj} = 2.73$, $N_{eff} = 19$, $P = 0.014$) (Fig. 3c). They were strongly correlated ($R^2 = 0.63$), indicating that they were both governed by the same factors.

The increasing breeding success could be explained by a gradual increase in the proportion of females successfully rearing chicks, rather than due to larger broods: in CAP the frequency of brood-less females decreased significantly ($b = -0.614$, $t = -2.95$, $P = 0.006$, $N = 30$), with mean proportions of 66.5% and 57.9% during 1979–1989 and 1991–2008, respectively. Brood sizes did not change ($b = 0.014$, $t = 1.50$, $P = 0.14$, $N = 30$), and averaged 2.56 and 2.65 chicks/brood in the two periods, respectively. A similar pattern was observed in BG; the proportion of brood-less females decreased significantly ($b = -0.618$, $t = -2.22$, $P = 0.035$, $N = 30$), and brood sizes only tended to increase ($b = 0.089$, $t = 1.75$, $P = 0.091$, $N = 30$), with 3.16 and 3.42 chicks/brood in the two periods, respectively.

4.3.1. Modifying factors – recruitment

Minimum temperatures in June 5–20 increased by 1.3 °C during the 30-year period (Table 2), and in both species annual variations in breeding success clearly correlated positively with this weather variable ($R_{CAP}^2 = 0.27$, $P = 0.003$; $R_{BG}^2 = 0.24$, $P = 0.006$; $N = 30$). There was no trend in the amount of rain in June, and apparently it did not influence breeding success ($R_{CAP}^2 < 0.001$, $R_{BG}^2 = 0.003$; $N = 30$). For other weather factors (date of snow melt in spring and April–May temperatures) correlations with breeding success were negligible, even though pre-laying April temperature increased by 2.5 °C and the date of snow-free ground advanced by 3 weeks (Table 2).

In the 1980s, an initially dense population of red fox was reduced nearly five-fold by the sarcoptic mange disease. Subsequently, the fox population fluctuated slightly, but at a lower level than before

Table 3

Mortality (%) and proportional predator loss of radio-marked adult^a male (M) and female (F) capercaillie and black grouse in early and late part of the 30-year monitoring period at Varaldskogen, Norway. Number of birds in (·).

	Capercaillie				Black grouse			
	1980–1990		1996–2003		1981–1982		2005–2008	
	M (69)	F (50)	M (22)	F (39)	M (9)	F (24)	M (36)	F (25)
Annual mortality	33	36	35	43	n/a	n/a	66	57
Seasonal monthly mortality ^b								
May + June ^b	2.5	3.2	4.3	6.1	0.0	4.2	17.9	4.1
Spring	4.4	3.4	4.4	4.4	–	–	11.9	4.8
Summer	1.8	1.7	1.9	2.0	–	–	5.2	2.3
Autumn	4.4	0.9	4.2	1.0	–	–	6.6	5.1
Winter	5.2	5.4	4.6	3.8	–	–	8.8	9.6
Proportions (%) ^c								
Goshawk	29	61	31	63	–	–	64	50
Red fox	40	13	49	17	–	–	19	23
Pine marten	25	23	12	17	–	–	13	24
Unknown	6	3	8	3	–	–	4	3

Source: Wegge et al. (1988, 1992), unpubl. (CAP), Kolstad et al. (1985), and Wegge and Rolstad unpubl. (BG).

^a Birds > 11 months of age.

^b Spring = April and May, Summer = June–August, Autumn = September–November, Winter = December–March. Black grouse data in 1980s only from May to June.

^c Proportions of total annual predation.

(Fig. 5). In both species, annual breeding success correlated negatively with an index of fox abundance based on yearly hunting bags ($R^2_{CAP} = 0.18$, $P = 0.02$; $R^2_{BG} = 0.30$, $P = 0.002$; $N = 30$). Combining June minimum temperature and fox abundance in a multiple regression showed that they both contributed significantly and about equally – in opposite directions – in explaining breeding success in each species (CAP: $r_{JuneT} = 0.42$, $P = 0.03$, $r_{fox} = -0.43$, $P = 0.02$; BG: $r_{JuneT} = 0.49$, $P = 0.006$, $r_{fox} = -0.43$, $P = 0.02$). Nevertheless, with breeding success increasing and numbers of adult BG decreasing, or remaining more or less stable as in CAP, annual recruitment of young birds in autumn was not sufficient to balance post-autumn losses among fledged chicks plus older birds in either of the two species.

4.4. Habitat use

Habitat studies conducted in the 1980s showed expected patterns: Adult CAP was closely associated with old, semi-natural forest during all seasons (Fig. 6a and b). During spring and summer, BG used younger seral stages (Fig. 6c and d). In both species, nesting

sites were distributed on all seral stages, with no preferences for any one age class of forest (Storaas and Wegge, 1987, unpubl.).

The distribution of adult birds on forest age classes changed in later years, with more use of plantations by CAP. However, due to the concurrent change in availability (old forest cover decreasing and plantations increasing), the new distributions did not signify changes in preferential use, except in the case of females and broods (Fig. 6a and Table 4): In later years, the oldest plantations received proportionally more use and old, semi-natural forest less use by females. Similarly, in the 1980s, CAP broods were mainly found in old, *Vaccinium*-rich semi-natural forest, whereas in the late 1990s and early 2000s they used middle-aged plantations extensively and older plantations preferentially (Table 4). In the early 1980s, BG broods were also more associated with old forests than with young plantations, then mainly using pine-forested bogs. In 2007, this pattern had changed: BG broods were now using young and middle-aged plantations much more frequently, although not preferentially because plantations covered a much larger proportion of the area (Table 4). Selected plantations usually had a good coverage of ericaceous shrubs, especially bilberry.

In the total 100 km² study area in early 1980s, only one of 21 CAP leks (5%) was located in middle-aged plantations. After 2000, within a smaller 50 km² area, 4 of 12 leks (33%) were located in such forest ($P = 0.05$, FET).

5. Discussion

5.1. Population trends

Both species fluctuated in abundance during the 30-year period, but the general long-term trends were clear. In spite of a significant loss and fragmentation of old forest, the abundance of adult CAP did not decrease and that of adult BG did not increase, but decreased. Also, breeding success of both species, expressed as number of chicks per female in August, increased. Apparently, the latter trend was caused mainly by a larger proportion of females rearing young, indicating less egg predation in later years. Red fox was the main predator on grouse nests in the study area (Storaas and Wegge, 1987). Following a slight recovery from the sarcoptic mange disease in the early 1990s, the abundance of this predator declined during the last 10–15 years (Fig. 5), presumably due to a reduced food base caused by lower vole population peaks, closing of local garbage dumps, and less offal remains from moose hunting

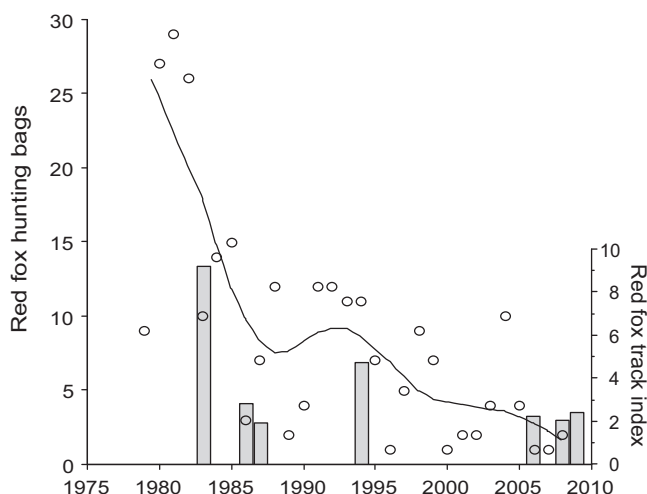


Fig. 5. Number of red foxes harvested during 1979–2008 within the 80 km² Varald State Forest (open circles with best fitted trend line; lowess, tension = 30%), and number of red fox tracks per 10 km per 24 h after end of snow fall within the main study area of the Varald State Forest (bars).

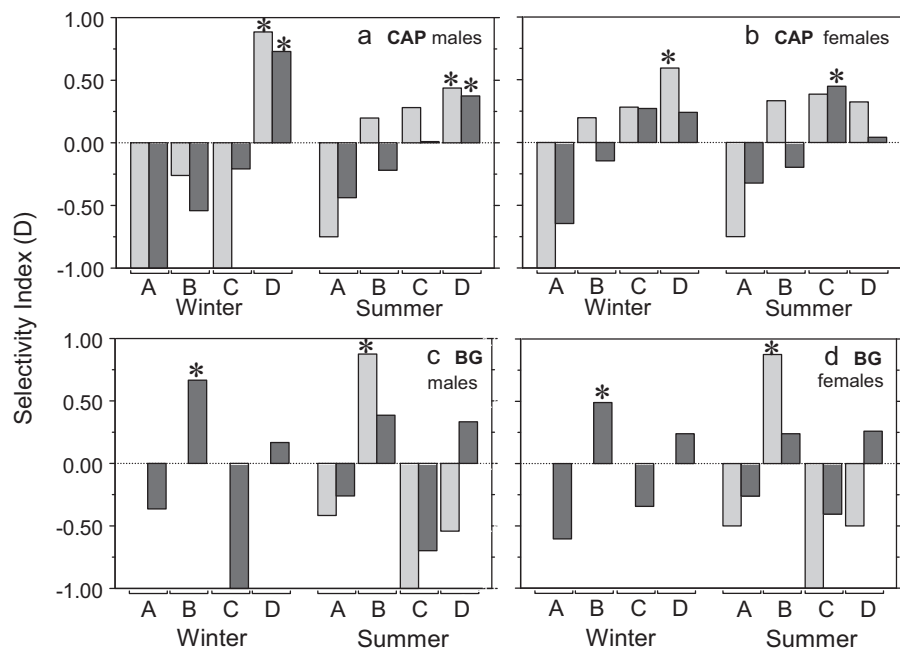


Fig. 6. Distribution of radio-marked capercaillie (CAP) and black grouse (BG) on forest age classes during December–April (winter) and May–November (summer) in the early (1979–1986, grey) and late (2000–2008, black) part of the monitoring period, illustrated by Ivlev's selectivity index. Logged and regenerated stands: A = 0–20 yrs, B = 21–40 yrs, C ≥ 40 yrs. D = old, semi-natural forest. * denotes preferential use ($P < 0.05$) based on Bonferroni adjusted Chi-Square tests.

Source: Rolstad (1988), Rolstad et al. (1988), Wegge (1985), Rolstad and Wegge (1988), Saxebøl (2000), Breisjøberget (2001), Haagensen (2003), and unpubl. (CAP, Kolstad et al. (1985), and Wegge and Rolstad unpubl. (BG)).

(unpubl.). During the monitoring period, annual losses estimated from the sampling in August. (CAP males/females: 34/39%, and BG m/f: 53/46%) appeared – in general – to be balanced by local recruitment. Exceptions were noted in 1985 (CAP) and in 1992 and 1993 (BG), when losses were estimated at less than 10%, indicating some immigration. For BG this did not affect the population trend line since immigration was indicated to occur in the middle of the monitoring period. In CAP, the trend lines would have been slightly more positive for males and less negative for females if the suspected early-period immigration had not occurred. Emigration was minimal or nil, as in the few years when annual losses were much higher than average, autumn recruitment was also quite high; overwinter mortality among juveniles is known to exceed that of adults. Also, adult birds that were radio-marked within the study area did not disperse out of the area, except for a few that wintered outside but returned in spring (Rolstad et al., 1991, unpubl.). Among a small sample of CAP chicks that were radio-marked in early autumn, males remained within the study area and females dispersed without returning (unpubl.). With populations assumed to be demographically closed, the increasing recruitment of young birds in autumn was not sufficient to offset periods of high local losses among older birds. This pattern was most pronounced in BG.

In CAP, females and males declined and increased only slightly, respectively, combined leading to a lower female-biased sex ratio.

Previous studies (Wegge, 1980; Lindén, 1981; Moss and Oswald, 1985) have shown that in years of poor breeding, more female than male chicks are recruited. Wegge (1980) showed that this is due to a higher mortality of the male sex within the broods. In the present study, females outnumbered males by a factor of nearly 2 among 323 chicks that were classified to sex during the August censuses (207 and 116, respectively), but brood sizes did not change much during the 30-year period. In the female-biased population with a low density of adult males, the rather low number of male recruits was sufficient to balance the losses among older males. Since most female chicks disperse from the study area, whereas male chicks are recruited locally, the declining trend among older females is more difficult to explain. It could possibly be due to a combination of more chicks leaving the study area than receiving from outside (source-sink dynamics), or a higher local predation pressure on female CAP in later years. Population declines of boreal forest grouse have almost invariably been attributed to reduced recruitment of young birds (e.g. Storch, 2007; Moss et al., 2001; Kurki et al., 1997, 2000; Ludwig et al., 2008; Summers, in press); only a few studies have indicated that the growth rate may be more affected by adult survival (Rajala, 1974; Caizergues and Ellison, 1997; Kauhala and Helle, 2002). In our study area, trends in adult numbers declined (BG) or remained more or less constant (CAP) during the 30-year period, in spite of increasing recruitment rates.

Table 4
Distribution of capercaillie and black grouse broods (%) on forest age classes in early (1980s) and late parts (1999–2007) of the 30 year monitoring period. Sample sizes in (·), indicated by number of broods/number of habitat locations. Bold figures denote statistical significant preference (Chi square with Bonferroni adjustment of alpha, $P < 0.05$).

Age class (years)	Proportions Early/late	Capercaillie		Black grouse	
		Early (27/741)	Late (10/287)	Early (4/172)	Late (6/124)
Plantations					
0–20	38.0/18.1	14	2	13	7
21–40	8.6/28.9	17	30	30	30
>40 yrs	4.6/22.3	7	44	5	29
Old, semi-natural forest >70 yrs	48.8/30.7	62	24	66	34

Source: Rodem et al. (1984), Bodal (2001), Lehn (2002), Wegge et al. (1992, 2007), unpubl. (CAP), Kolstad et al. (1985) and Wegge and Rolstad unpubl. (BG).

Hence, and like inferred in the latter studies, our study populations appeared to be regulated more by the mortality of adults and fledged chicks than by breeding success.

5.2. Weather and predation

During the course of the study, spring temperatures increased steadily, and on average, the ground was snow-free three weeks earlier by the end of the study period. Ludwig et al. (2006) predicted that earlier springs might adversely affect the breeding performance of BG due to a mismatch between earlier date of mating – and subsequent birth of chicks – and availability of food for the young chicks. This prediction was not supported by our data at Varaldskogen. Instead, the opposite trend was observed: despite a general increase in spring temperatures, breeding success did not decrease but increased. Furthermore, in 2007 April temperatures were especially high with peak of mating occurring about one week earlier than “normal”, the most pronounced advancement ever recorded (Wegge et al., 2010). That year, both CAP and BG females reared more chicks than average (Fig. 3c).

Although summer temperature did not increase nearly as much as that in winter and spring, the minimum temperature during the first 2 weeks after hatching in June increased by 1.3 °C, and year-to-year variation in breeding success correlated significantly with this temperature variable. Number of surviving chicks in the broods tended to increase during the course of the study. Hence, in addition to lower egg loss from fewer foxes, the positive trend in recruitment might have been facilitated by better conditions for the young chicks, mediated by a more favourable temperature regime after hatching. The decreasing trends in adult abundances (BG both sexes and CAP females) coincided with general trends of less snow cover, milder winters and earlier snow-melt. Both species are adapted to roosting in snow. Like reported from a study of BG elsewhere in Norway during a year with little snow (Spidsø et al., 1997), lack of suitable snow conditions might have exposed the birds – especially males – to high goshawk predation during winter. However, in our study adult mortality in black grouse was not correlated with winter weather conditions. Instead, the high predation loss by goshawk in later years may have been due to more goshawks wintering in the study area, combined with more favourable habitat conditions for hunting (below). Yearly counts of migrating goshawks at the Falsterbo bird station in southern Sweden – presumably an indicator of winter “floaters” – increased steadily from the mid-1970s up till about 1990, and stayed twice as high after 1990 compared to the period before (median_{1975–1990} = 17, median_{1991–2008} = 35; Mann–Whitney $U=220$, $Z=2.20$, $P=0.03$, $N_1=16$, $N_2=18$) (Falsterbo Bird Observatory, 2010). Although not systematically recorded, we also observed goshawks more frequently during fieldwork in later years. Excluding 3 years in the early 1990s, when we suspected immigration of BG to Varaldskogen, losses increased significantly with increasing number of migrating goshawks, most markedly in BG males ($R^2=0.40$, $P<0.001$, $N=26$, Fig. 7), less so in BG females ($R^2=0.26$, $P=0.008$, $N=26$). No relationships were found in CAP males and females.

5.3. Habitat use

With the conversion of old forest to clearcuts and plantations, habitat quality was expected to have become poorer for CAP and better for BG, leading to declining and increasing numbers of the two species, respectively. This did not happen. The loss and fragmenting of old forest into small patches did not adversely affect CAP mating performance, as several new leks were established in middle-aged plantations (Rolstad et al., 2007). The increased proportion of middle-aged plantations did not seem to have affected the BG negatively, like has been speculated from census work in

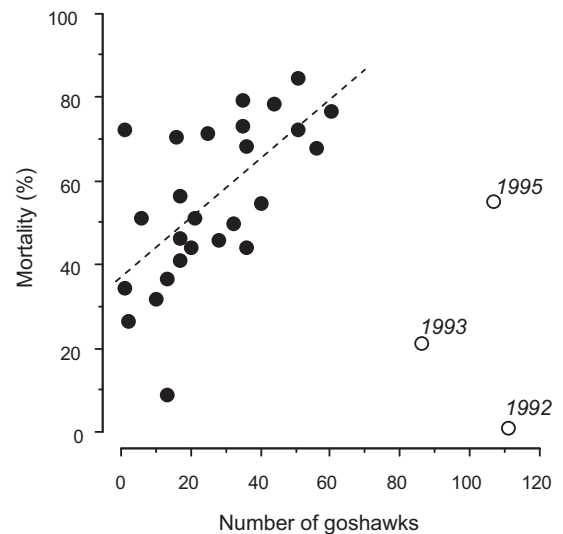


Fig. 7. Relationship between mortality of BG males at Varaldskogen and number of migrating goshawks observed at the Falsterbo Bird Station in southern Sweden (Falsterbo Bird Observatory, 2010). The three 1990-years were assumed outliers, due to immigration, and were excluded from the correlation analysis. See text for more details.

Finland (Helle and Helle, 1991). We recorded no change in the relative use of such regeneration stands during spring and summer by adult BG between the early 1980s and the last three years. While BG broods used edges of old pine-forested bogs preferentially in the early years (Kolstad et al., 1985; Wegge and Kastdalen, 2008), in later years they were more frequently located in young and middle-aged pine and mixed pine/spruce plantations.

Young grouse chicks require protein-rich animal food for growth and survival, of which Lepidopteran larvae are a main component (Picozzi et al., 1999; Wegge and Kastdalen, 2008). This food is associated mainly with bilberry shrubs (Atlegrim and Sjöberg, 1995; Stuen and Spidsø, 1988; Picozzi et al., 1999; Wegge et al., 2005), which is reduced after clearcutting (Atlegrim and Sjöberg, 1996a,b; Kardell, 1980; Uotila and Kouki, 2005; Uotila et al., 2005; Kvasnes and Storaas, 2007). Thus, plantations have long been assumed to be unsuitable habitats for broods (Kastdalen and Wegge, 1985; Moss and Oswald, 1985; Storch, 2007). From Finland, however, Lakka and Kouki (2009) recently reported that bilberry and associated insects – following their near eradication in the young sapling stage – had recovered in middle-aged plantations (30–40 years) to the same level as in mature, natural forests. In the late 1990s and early 2000s, our radio-marked broods of both species were commonly located in such middle-aged plantations. Although the abundance of insect larvae was lower in nearby control sites compared to where the broods were, small openings with bilberry and associated larvae provided foraging sites with surrounding denser forest providing cover against goshawk predation (Lehn, 2002, unpubl.).

Furthermore, CAP broods may have been more vulnerable to predation during the first part of the study period. The landscape at that time had a higher contrast, with remnant old forests embedded in a matrix of new clearcuts and very young plantations (Fig. 4). This might have led broods to restrict their movements to remnant strips and patches of old forest (Storaas et al., 1999) or force them to cross unsuitable habitat, both exposing them to high mortality (Wegge et al., 1990). As plantations grew older, with a re-established field layer of bilberry, they became more attractive brood habitats. If this holds true, the broods might have become more spread out in the total forest landscape and therefore less predictable prey. Concurrently, plantations became denser, thereby precluding efficient hunting by goshawks.

On a large landscape scale in Finland, Kurki et al. (2000) reported a significant correlation between a decreasing breeding success of CAP and BG and an increasing percentage of agricultural land and a decreasing percentage of older forests at the cost of young forests. They inferred that this was related to lower breeding success due to a positive numerical response in predators. Although this may hold true regarding the influence of agricultural land, our results do not support the latter finding, as breeding success in CAP did not decrease with increasing loss and fragmentation of old forest. Instead, our results concur more with recent reports from Finland: a steep decline of CAP during 1965–1988 could not be explained by a proportional increase in younger forest (Sirkiä et al., 2010; Miettinen, 2009), on the basis of analyses of time-series data on the landscape scale, concluded that the long-term decline of CAP had stopped in the 1990s. He and co-workers (Miettinen et al., 2010) reported extensive use of middle-aged pine and spruce plantations by both adult birds and broods – a result very much in agreement with our findings.

In our study area in the early 1990s, forestry started a practice of thinning the multi-layered old, pine-dominated forests and subsequently fertilizing them in order to stimulate a growth-spurt of the residual trees before final harvesting. This practice removed most of the spruce trees in the lower strata besides thinning the upper layer. The remaining old forest stands became far more open not only from above, but also in the vertical plane. With dwarf shrubs and patchy regenerating spruce in the lowest stratum, these open, old forests appeared to be attractive habitat for adult BG throughout the year. However, they may have been “ecological traps” (Battin, 2004). The park-like structure probably facilitated hunting by goshawks; not only did they provide optimal lookout sites but also facilitated the raptor during pursuits. From North America, Gullion and Alm (1983) reported a similar raptor-facilitation on ruffed grouse *Bonasa umbellus* when middle-aged forest was opened up by thinning. The fact that goshawks killed a larger proportion of male BG, and that this sex suffered higher mortality than females, supports this interpretation. In Finland, goshawk was found to have adapted well to a fragmented landscape consisting of a mosaic of different-aged plantations and remnant old forest (Tornberg and Colpaert, 2001); the birds frequently hunted in the ecotones between clearcuts and closed forest. Although prey density was lower there, hunting efficiency appeared to be higher in small patches than in closed-canopy, larger tracts (Tornberg and Sulkava, 1991).

Summing up, during a period of 30 years, when the forest landscape changed radically both in age composition and in fragmentation pattern, our grouse populations did not change much. Instead, they appeared to be modified and affected more by numerical and functional changes in the predator regime. In closing, however, we acknowledge the fact that our findings are based on a “case study”. Thus, the generality of the results should be treated cautiously and hopefully be scrutinized and tested in future studies.

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